



PRIME NUMBERS

THE MAGIC NUMBER FILTER

Discovering Prime Numbers with the
Sieve of Eratosthenes

THE MYSTERY OF SPECIAL NUMBERS

Did you know some numbers are special because they have only 2 factors? Let's be Math Detectives and find out! Which numbers have this magic power?



PRIME VS COMPOSITE

Prime = exactly 2 factors (1 and itself)

Composite = more than 2 factors

1 is special — not prime or composite!





THE SIEVE OF ERATOSTHENES - OUR MAGIC NUMBER FILTER

Imagine using a (tea sieve) to filter tea leaves – the Sieve of Eratosthenes filters numbers to find only the pure primes! We cross out multiples to find the magic soloists!

NUMBER GRID 1 TO 50 - LET'S GET READY!

Here is our playground of numbers. Let's start filtering!



1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

STEP 1 - CROSS OUT MULTIPLES OF 2

2 is the smallest prime and the only even prime number! Let's highlight it in bright yellow. Now, cross out all multiples of 2 (4, 6, 8, 10, 12...) in grey. These numbers can form equal teams, so they're not prime!



STEP 2 - CROSS OUT MULTIPLES OF 3

Next, 3 is a prime number! We highlight it in bright yellow and cross out all its multiples: 6, 9, 12, 15, 18, 21... Watch as more composite numbers get filtered out!



STEP 3 - CROSS OUT MULTIPLES OF 5 AND BEYOND

Now 5 is our next prime hero! Cross out all multiples of 5 like 10, 15, 20, 25... Then continue with 7 and cross out its multiples too. Keep crossing out multiples of primes to find the magical soloists!



THE PRIME TREASURE MAP

Look at the magic numbers left shining! These are all the prime numbers between 1 and 50.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

INTERACTIVE PRACTICE - IS IT PRIME?

Let's play "Whack-a-Prime!" Can you spot the primes?

11 → Prime or Composite?

13 → Prime or Composite?

15 → Prime or Composite?



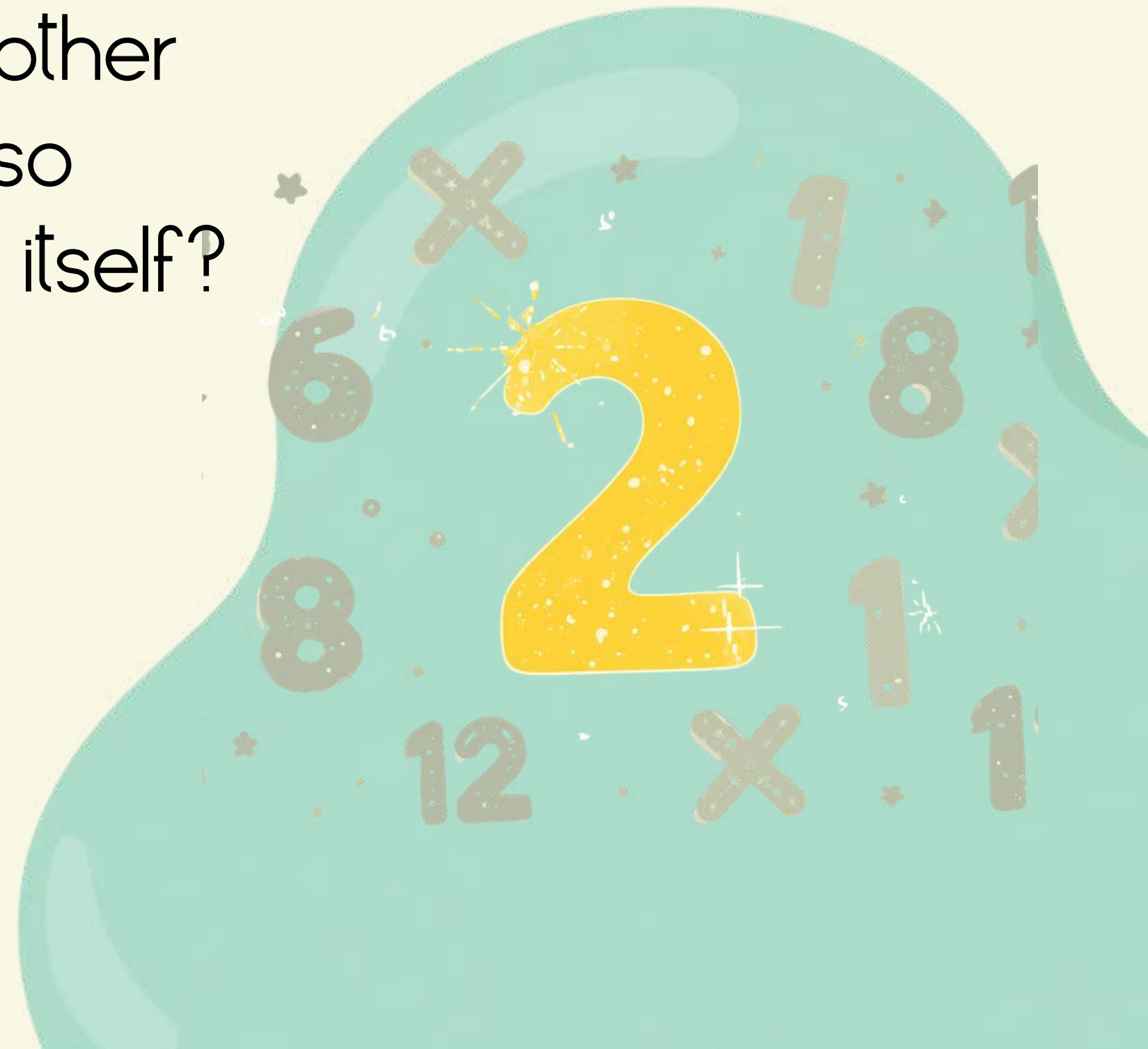
WHY IS 1 SPECIAL?

Number 1 is neither prime nor composite because it has only one factor — itself! Think of it this way: 1 child alone is not a team or a soloist — just unique!



REMEMBER - 2 IS SPECIAL TOO!

2 is the only even prime number! All other even numbers can be divided by 2, so they have more than 2 factors. But 2 itself? It's a magical soloist!



WHAT WE LEARNED TODAY

Prime numbers have exactly 2 factors

Composite numbers have more than 2 factors

The Sieve of Eratosthenes helps us find primes by crossing out multiples

Number 1 is unique — neither prime nor composite

2 is the only even prime number



THANK YOU, MATH DETECTIVES!

Great job today! Keep spotting the magic numbers all around you. See you in Next Session for more math adventures!

